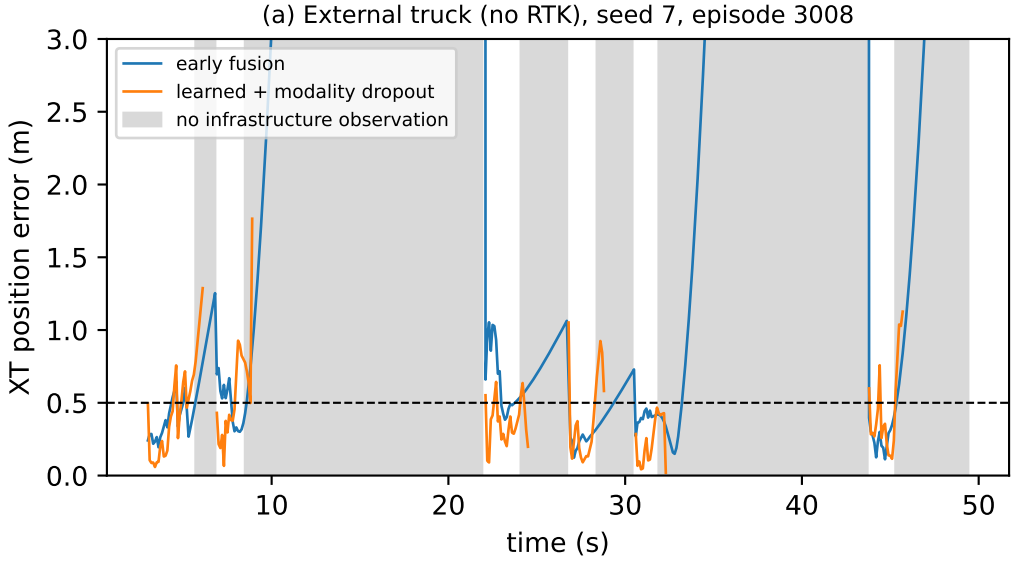
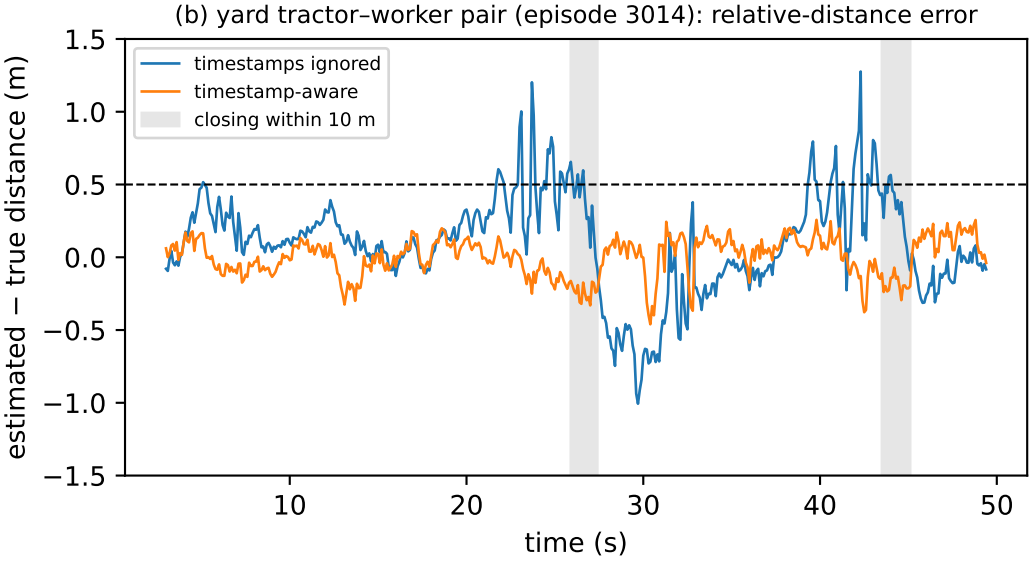

(a) External truck (no RTK), seed 7, episode 3008
early fusion
learned + modality dropout
no infrastructure observation
XT position error (m)
time (s)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
10
20
30
40
50


(b) yard tractor–worker pair (episode 3014): relative-distance error
timestamps ignored
timestamp-aware
closing within 10 m
estimated − true distance (m)
time (s)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
10
20
30
40
50